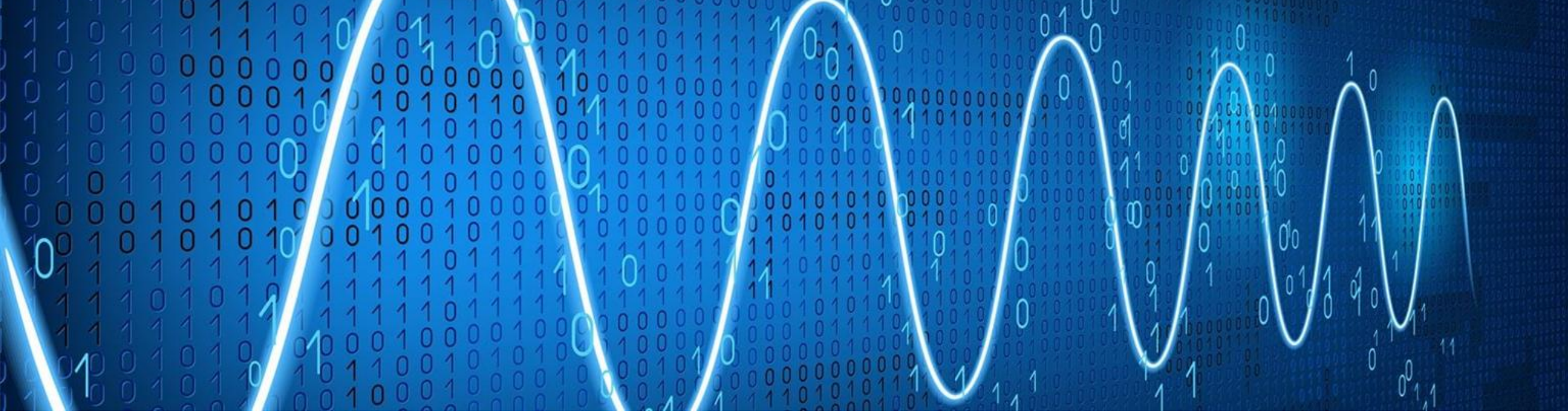




Lecture 12:

Fourier Transforms

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Where We Are in the Course

- Analog Signal Processing, Introduction to digital signal processing (Lectures 2-6)
- LTI System (Lectures 7 and 9)
- Z-Transform, System Function, Frequency Response (Lectures 8)
- FIR Filters (Lecture 8)
- IIR Filters, Linear Phase Filters (Lecture 10)
- Implementation Aspects (Lecture 11)
- ***CTFT, DTFT, DFT (Lecture 12)***
- FFT (Lecture 13)

Lecture 14: Sample Questions

Outline

- **Introduction: The Power of the Frequency Domain**

Why Fourier Transforms Matter in Signal Processing

- ***Continuous-Time Fourier Transform (CTFT)***

Continuous Time $\leftrightarrow$ Continuous Frequency

$$x(t) \xleftrightarrow{\text{CTFT}} X(j\omega)$$

- ***Discrete-Time Fourier Transform (DTFT)***

Discrete Time $\leftrightarrow$ Continuous Frequency

$$x[n] = x(nT_s) \xleftrightarrow{\text{DTFT}} X(e^{j\omega})$$

- ***Discrete-Fourier Transform (DFT)***

Discrete Time $\leftrightarrow$ Discrete Frequency

$$x[n] \xleftrightarrow{\text{DFT}} X[k]$$

Sections 10 & 11, Signal Processing First

Introduction: The Power of the Frequency Domain

The Power of the Frequency Domain: Why Fourier Transforms Matter in Signal Processing

- Signals are often **complex** in the time domain, making it difficult to analyze patterns. The frequency domain provides a clearer view of the signal's characteristics, such as **dominant frequencies**, **harmonics**, and **noise**.



The Power of the Frequency Domain: Why Fourier Transforms Matter in Signal Processing

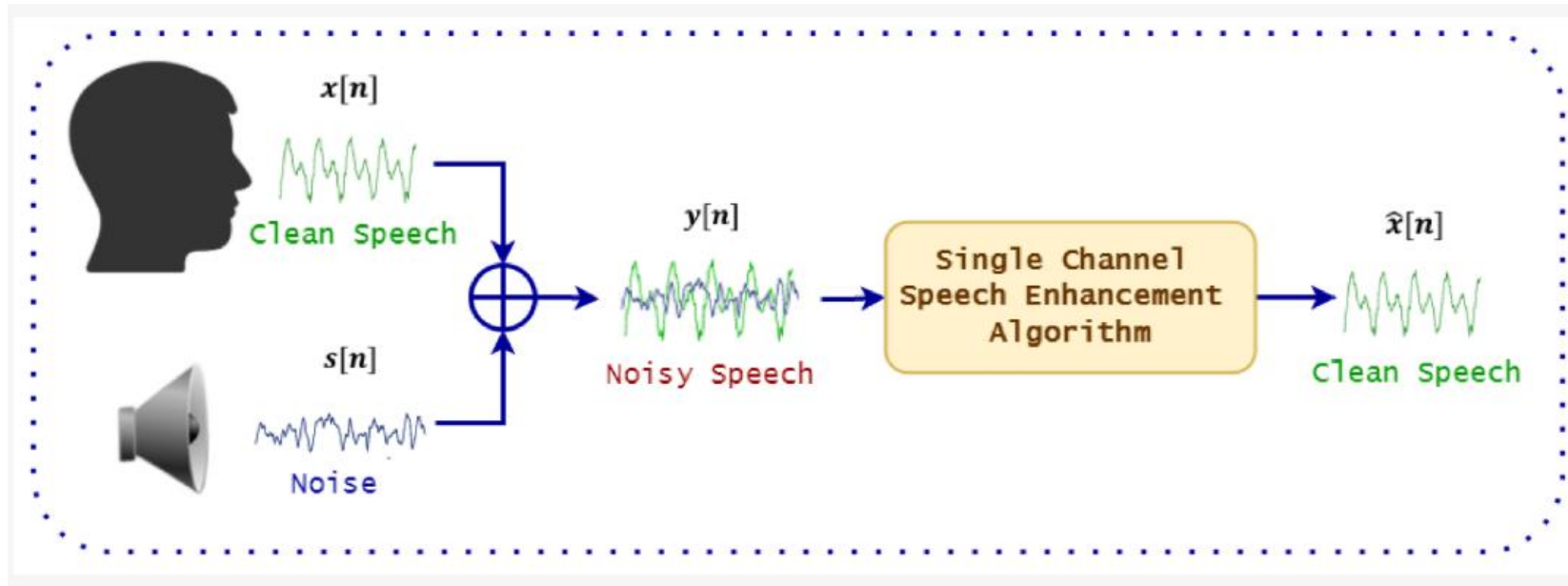
Fourier Transforms Provide insights into signal behaviors that are not obvious in the time domain.

Fourier Transforms Convert signals between time and frequency domains for better **analysis, filtering, compression, classification, parameter estimation, etc.**

Applications of Frequency-Domain Signal Processing

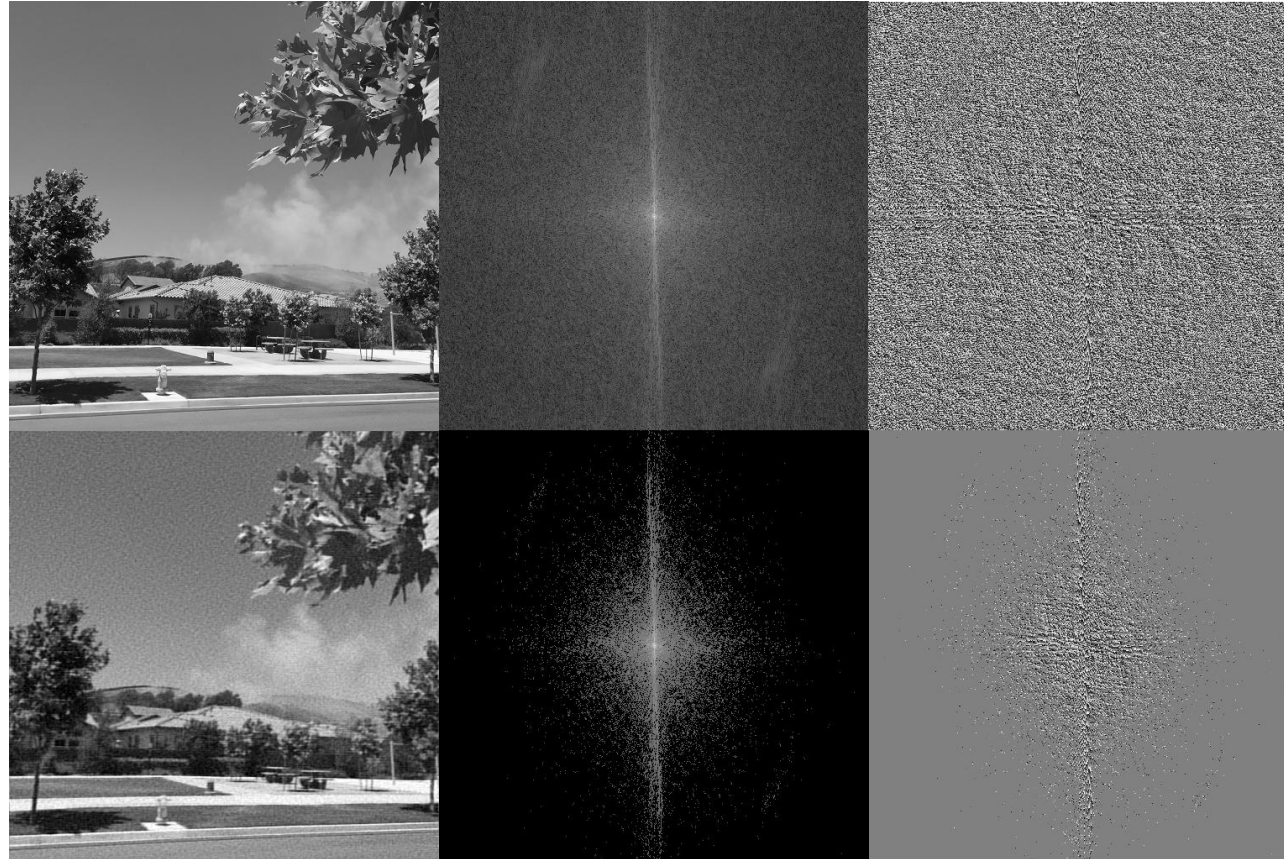
- **Audio & Speech Processing**
- **Image & Video Processing**
- **Medical Signal Processing**
- **Radar, Sonar & LiDAR**
- **Wireless Communication**
- **...**

Audio & Speech Processing: Noise reduction, music equalization, speech recognition, audio compression,



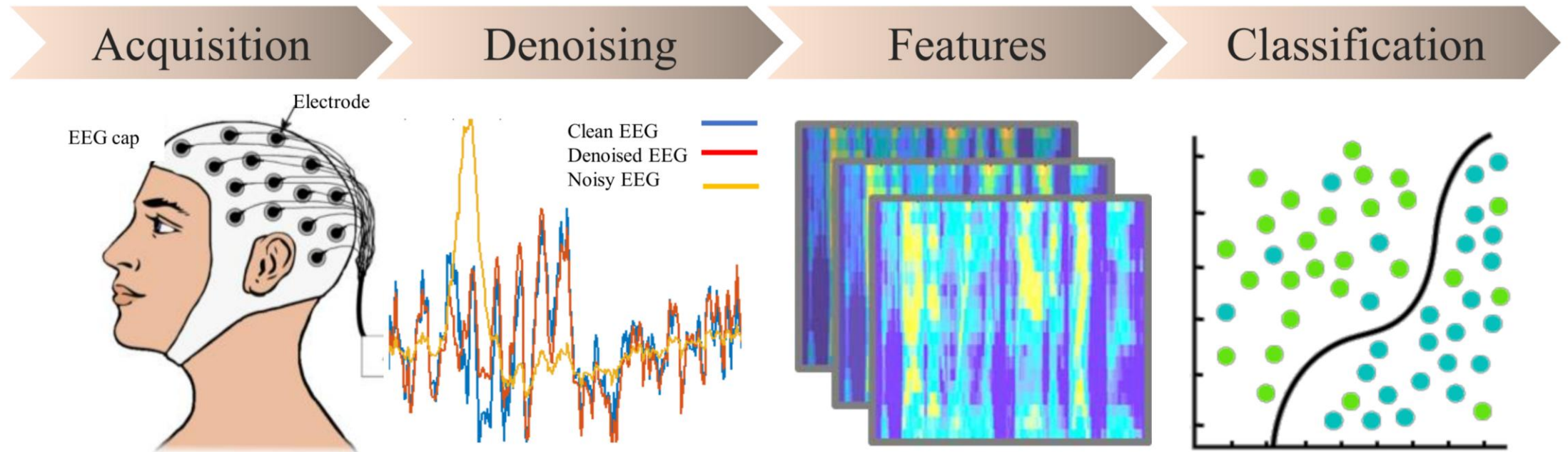
<https://www.mdpi.com/1424-8220/22/1/374>

Image & Video Processing: Compression (JPEG, MPEG), edge detection, pattern recognition, object detection, object classification , ...



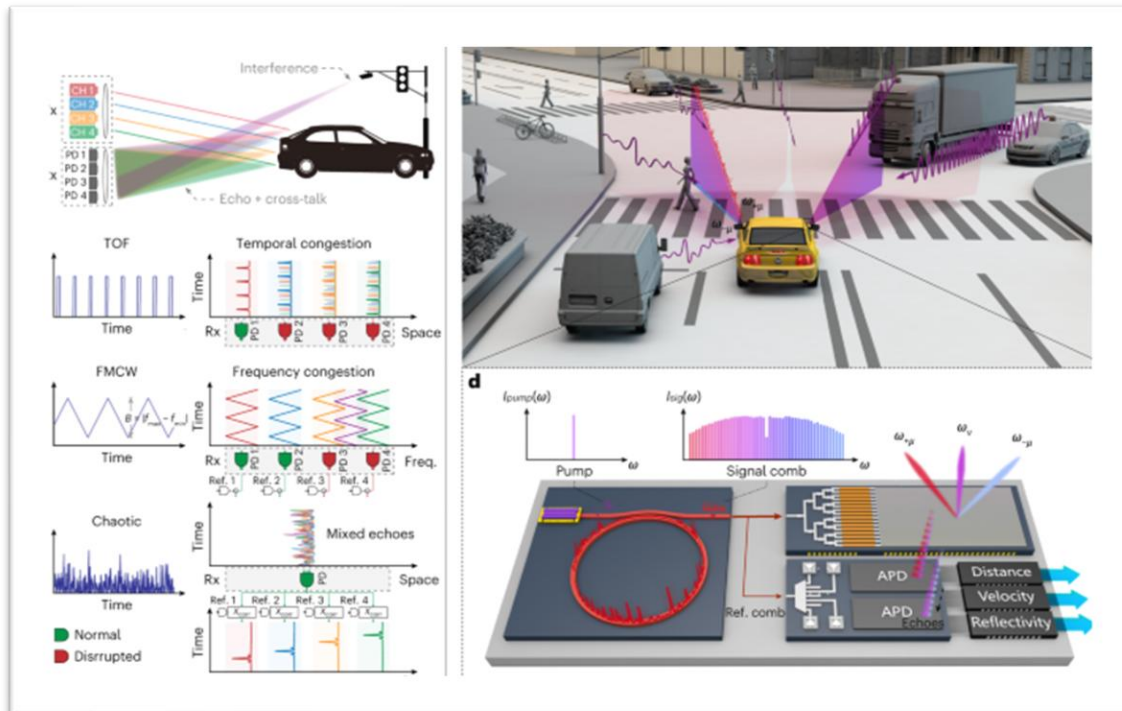
<https://blog.demofox.org/2020/11/04/frequency-domain-image-compression-and-filtering/>

Medical Signal Processing: MRI, ECG/EEG signal analysis, image-based medical signal processing (classifications and prediction)

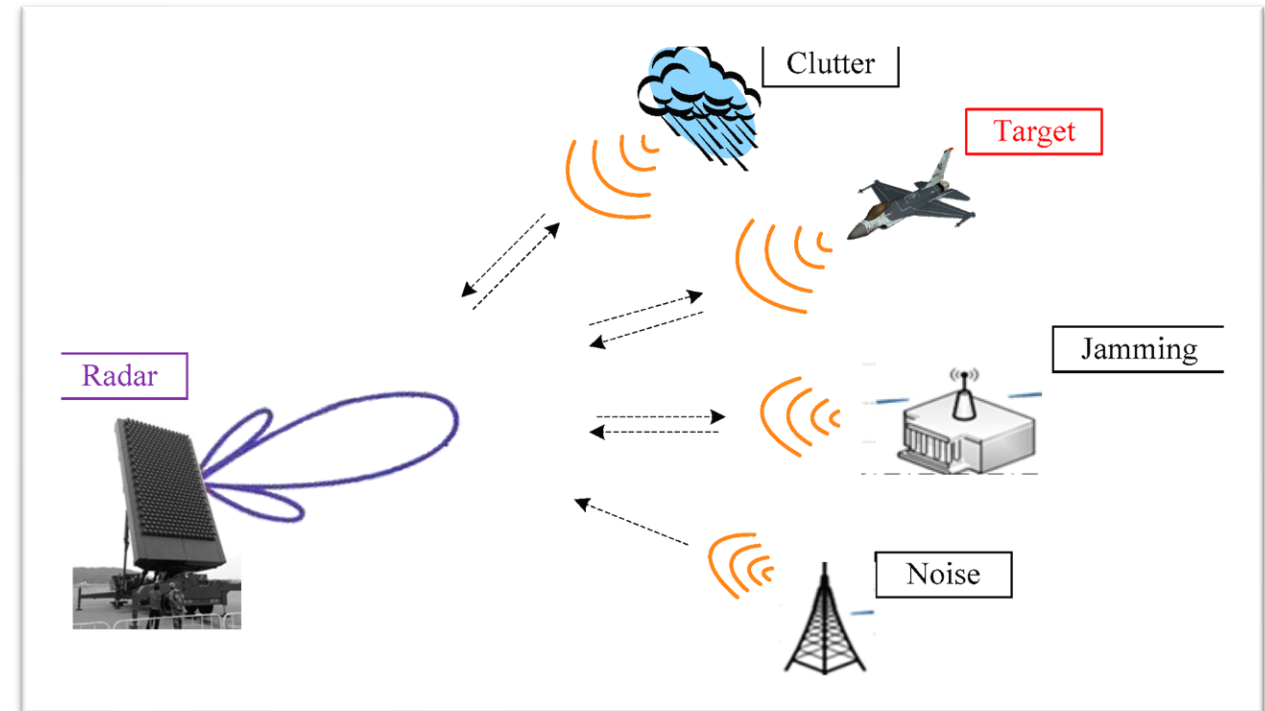


<https://www.mdpi.com/1424-8220/23/14/6434>

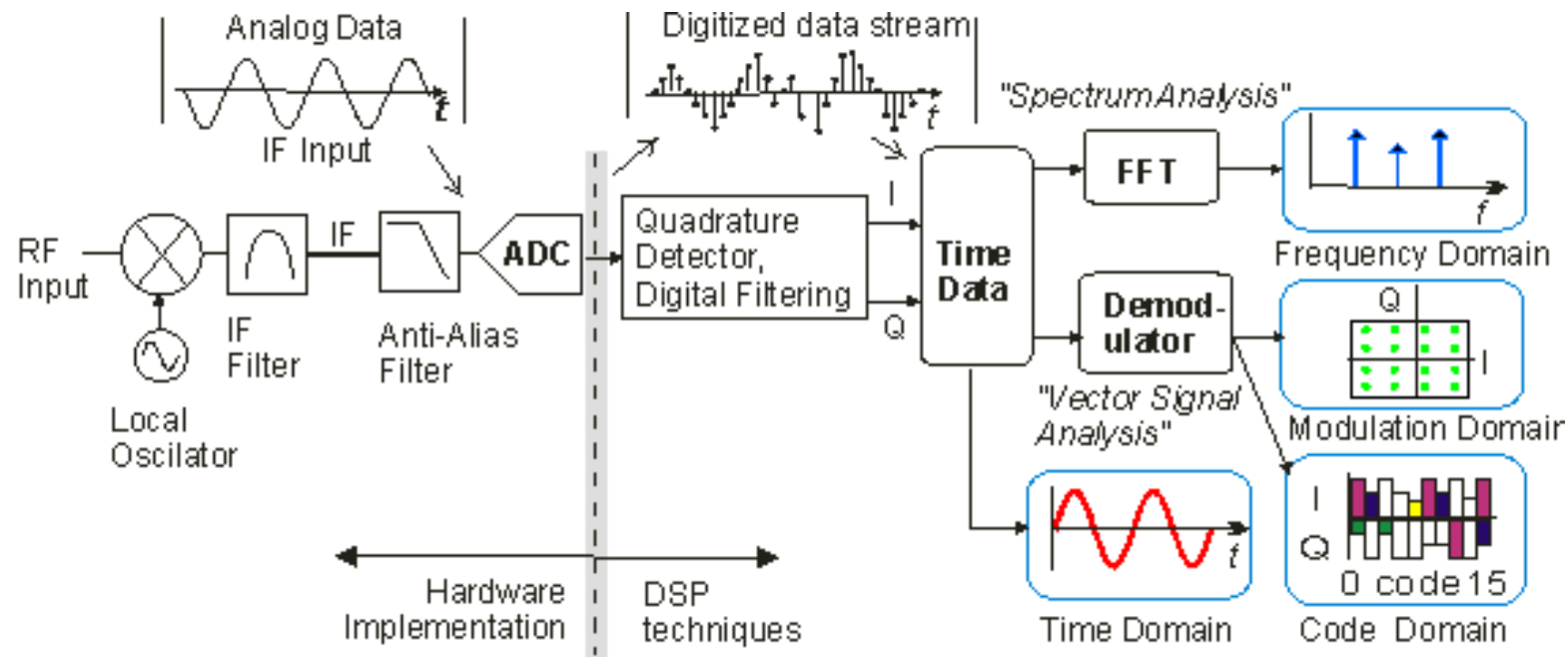
Radar, Sonar, Lidar: Object detection, classification, and parameter estimation



<https://www.nature.com/articles/s41566-023-01158-4>



Wireless Communication: Modulation, filtering, spectrum analysis.



VSA System Implementation - The Vector Signal Analyzer digitizes the analog input signal and uses Digital Signal Processing technology to process and provide data outputs; FFT algorithm produces frequency domain results, Demodulator algorithms produce modulation and code domain results.

Continuous-Time Fourier Transform (CTFT)

Continuous-Time Fourier Transform (CTFT)

Continuous Time $\leftrightarrow$ Continuous Frequency

$$x(t) \xleftrightarrow{\text{CTFT}} X(j\omega)$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

Inverse CTFT

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

Example of CTFT

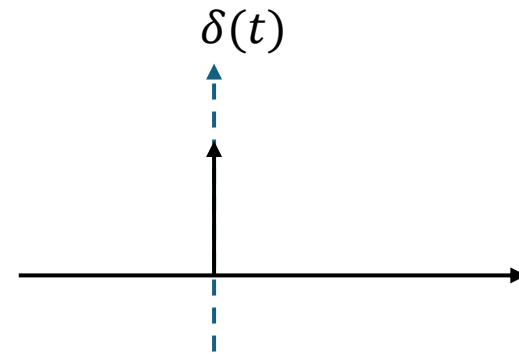
The Dirac delta is a signal that:

- Is zero everywhere except at $t = 0$
- Has infinite amplitude at $t = 0$
- Has unit area

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{+\infty} \delta(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{+\infty} \delta(t) e^{-j\omega \times 0} dt \\ &= \int_{-\infty}^{+\infty} \delta(t) dt = 1 \end{aligned}$$

$$\delta(t) \leftrightarrow 1$$



$$x(t) \delta(t - t_0) = x(t_0) \delta(t - t_0)$$

Example of CTFT

- If $X(j\omega) = 2\pi \delta(\omega - \omega_0)$, what is $x(t)$?

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} 2\pi \delta(\omega - \omega_0) e^{j\omega t} d\omega \\ &= \int_{-\infty}^{+\infty} \delta(\omega - \omega_0) e^{j\omega_0 t} d\omega \\ &= e^{j\omega_0 t} \int_{-\infty}^{+\infty} \delta(\omega - \omega_0) d\omega \\ &\quad e^{j\omega_0 t} \int_{-\infty}^{+\infty} \delta(\tilde{\omega}) d\tilde{\omega} \\ &= e^{j\omega_0 t} \end{aligned}$$

$$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$$

Properties of CTFT

- Linearity:

If

$$\begin{aligned}x_1(t) &\leftrightarrow X_1(j\omega) \\x_2(t) &\leftrightarrow X_2(j\omega)\end{aligned}$$

Then

$$a_1x_1(t) + a_2x_2(t) \leftrightarrow a_1X_1(j\omega) + a_2X_2(j\omega)$$

Properties of CTFT

- Delay:

$$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$$

- *Convolution*

$$x(t) * h(t) \leftrightarrow X(j\omega) H(j\omega)$$

*For LTI systems: $y(t) = x(t) * h(t)$, hence, $Y(j\omega) = X(j\omega) H(j\omega)$, where $H(j\omega)$ is the frequency response of the continuous-time system*

Properties of CTFT

Table 11-3: Basic Fourier transform properties.

Table of Fourier Transform Properties		
Property Name	Time-Domain: $x(t)$	Frequency-Domain: $X(j\omega)$
Linearity	$ax_1(t) + bx_2(t)$	$aX_1(j\omega) + bX_2(j\omega)$
Conjugation	$x^*(t)$	$X^*(-j\omega)$
Time-Reversal	$x(-t)$	$X(-j\omega)$
Scaling	$x(at)$	$\frac{1}{ a }X(j(\omega/a))$
Delay	$x(t - t_d)$	$e^{-j\omega t_d}X(j\omega)$
Modulation	$x(t)e^{j\omega_0 t}$	$X(j(\omega - \omega_0))$
Modulation	$x(t)\cos(\omega_0 t)$	$\frac{1}{2}X(j(\omega - \omega_0)) + \frac{1}{2}X(j(\omega + \omega_0))$
Differentiation	$\frac{d^k x(t)}{dt^k}$	$(j\omega)^k X(j\omega)$
Convolution	$x(t) * h(t)$	$X(j\omega)H(j\omega)$
Multiplication	$x(t)p(t)$	$\frac{1}{2\pi}X(j\omega) * P(j\omega)$

Discussion Time I

What are the CTFTs of $\cos(\omega_0 t)$ and $\sin(\omega_0 t)$?

$$\cos(\omega_0 t) = \frac{e^{-j\omega_0 t} + e^{j\omega_0 t}}{2}$$

$$\sin(\omega_0 t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

$$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$$



CTFT

Table 11-2: Basic Fourier transform pairs.

Table of Fourier Transform Pairs	
<i>Time-Domain: $x(t)$</i>	<i>Frequency-Domain: $X(j\omega)$</i>
$e^{-at}u(t) \quad (a > 0)$	$\frac{1}{a + j\omega}$
$e^{bt}u(-t) \quad (b > 0)$	$\frac{1}{b - j\omega}$
$u(t + \frac{1}{2}T) - u(t - \frac{1}{2}T)$	$\frac{\sin(\omega T/2)}{\omega/2}$
$\frac{\sin(\omega_b t)}{\pi t}$	$[u(\omega + \omega_b) - u(\omega - \omega_b)]$
$\delta(t)$	1
$\delta(t - t_d)$	$e^{-j\omega t_d}$
$u(t)$	$\pi \delta(\omega) + \frac{1}{j\omega}$
1	$2\pi \delta(\omega)$
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$
$A \cos(\omega_0 t + \phi)$	$\pi A e^{j\phi} \delta(\omega - \omega_0) + \pi A e^{-j\phi} \delta(\omega + \omega_0)$
$\cos(\omega_0 t)$	$\pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)$
$\sin(\omega_0 t)$	$-j\pi \delta(\omega - \omega_0) + j\pi \delta(\omega + \omega_0)$
$\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$	$\sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$
$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi}{T}k)$

Discrete-Time Fourier Transform (DTFT)

Discrete-Time Fourier Transform (DTFT)

Discrete Time $\leftrightarrow$ Continuous Frequency

$$x[n] = x(nT_s) \overset{\text{DTFT}}{\longleftrightarrow} X(e^{j\omega})$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}$$

**Periodic with
period of 2π**

Inverse DTFT

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{+\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Example of DTFT

For $x[n] = \delta[n]$:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{+\infty} \delta[n]e^{-j\omega n}$$

$$= \delta[0]e^{-j\omega \times 0}$$

$$= 1$$

$$\delta[n] \leftrightarrow 1$$

Example of DTFT

- If $X(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{+\infty} \delta(\omega - \omega_0 - 2\pi k)$, what is $x(t)$?

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\pi}^{+\pi} X(e^{j\omega}) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{+\pi} 2\pi \delta(\omega - \omega_0) e^{j\omega n} d\omega \\ &= \int_{-\pi}^{+\pi} \delta(\omega - \omega_0) e^{j\omega_0 n} d\omega \\ &= e^{j\omega_0 n} \int_{-\infty}^{+\infty} \delta(\omega - \omega_0) d\omega = e^{j\omega_0 n} \end{aligned}$$

$$e^{j\omega_0 n} \leftrightarrow 2\pi \sum_{k=-\infty}^{+\infty} \delta(\omega - \omega_0 - 2\pi k)$$

Properties of DTFT

- Linearity:

$$a_1 x_1[n] + a_2 x_2[n] \leftrightarrow a_1 X_1(e^{j\omega}) + a_2 X_2(e^{j\omega})$$

- Delay:

$$x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(e^{j\omega})$$

- Convolution

$$x[n] * h[n] \leftrightarrow X(e^{j\omega}) H(e^{j\omega})$$

*For LTI systems: $y[n] = x[n] * h[n]$, hence, $Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$ where $H(e^{j\omega})$ is the frequency response of the system*

Discussion Time II

If $x[n] = \cos(\omega_1 n) + \cos(\omega_2 n)$ passes through an ideal low-pass filter with cutoff frequency ω_c , where $\omega_1 < \omega_c < \omega_2$, what is the output?

$$\cos(\omega_0 n) = \frac{e^{-j\omega_0 n} + e^{j\omega_0 n}}{2}$$

$$e^{j\omega_0 n} \leftrightarrow 2\pi \sum_{k=-\infty}^{+\infty} \delta(\omega - \omega_0 - 2\pi k)$$



Discrete-Fourier Transform (DFT)

Discrete-Fourier Transform (DFT)

Discrete Time $\leftrightarrow$ Discrete Frequency

$$x[n] = x(nT_s) \overset{\text{DFT}}{\longleftrightarrow} X[k]$$

Why?

N-Point DFT:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} kn} \quad \text{for } k = 0, \dots, N-1$$

Inverse DFT

If the length of $x[n]$ is denoted by L and $L \leq N$:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j \frac{2\pi}{N} kn} \quad \text{for } n = 0, \dots, N-1$$

Discrete-Fourier Transform (DFT)

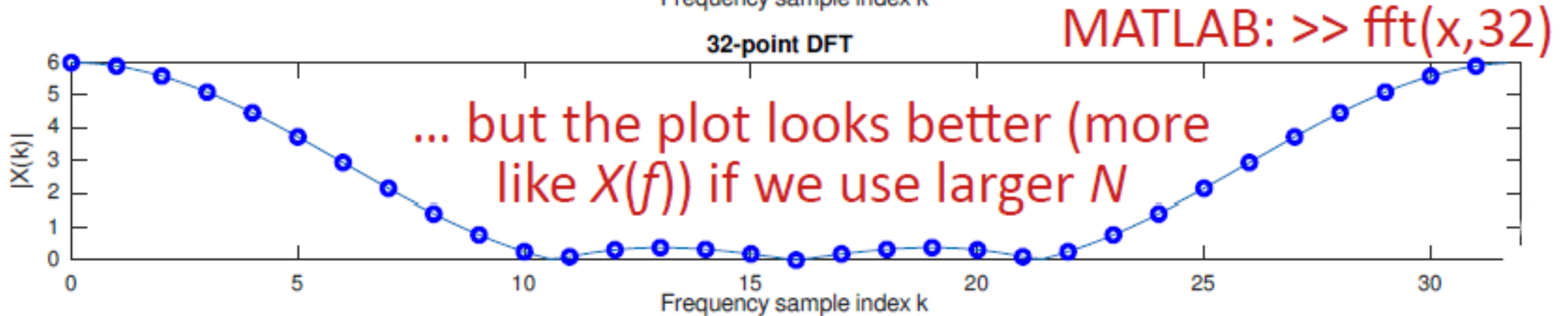
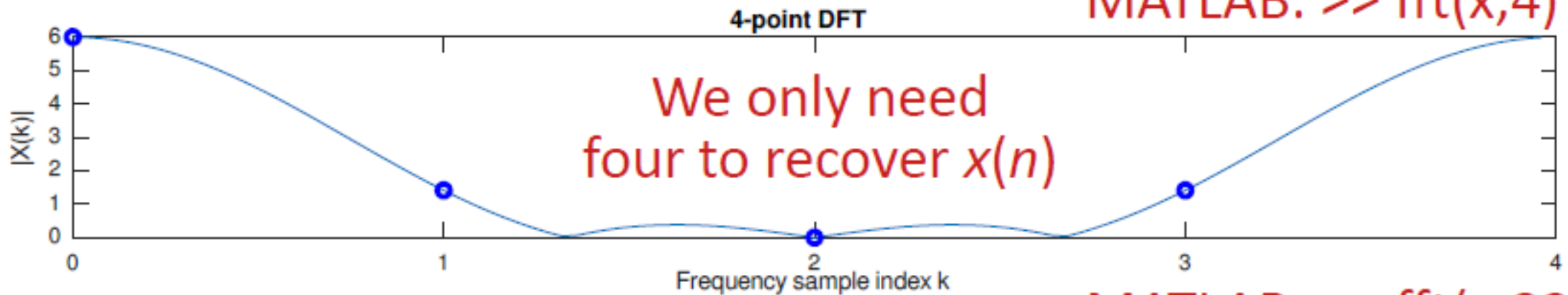
N-Point DFT:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} \quad \text{for } k = 0, \dots, N - 1$$

If the length of $x[n]$ is denoted by L and $L \leq N$:

$$X[k] = X(e^{j\omega}) \Big|_{\omega = \frac{2\pi}{N}k} = X\left(e^{j\frac{2\pi}{N}k}\right)$$

Example: $x(n) = [\underline{1} \ 2 \ 2 \ 1]$



Example of DFT

- $x[n] = [1 \ 2 \ 3]$, 3-Point DFT?

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} \quad \text{for } k = 0, \dots, N-1, N=3$$

$$X[k=0] = \sum_{n=0}^2 x[n] e^{-j\frac{2\pi}{3}0n} = 6$$

$$X[k=1] = \sum_{n=0}^2 x[n] e^{-j\frac{2\pi}{3}n} = -1.5 + j0.866$$

$$X[k=2] = \sum_{n=0}^2 x[n] e^{-j\frac{2\pi}{3}2n} = -1.5 - j0.866$$